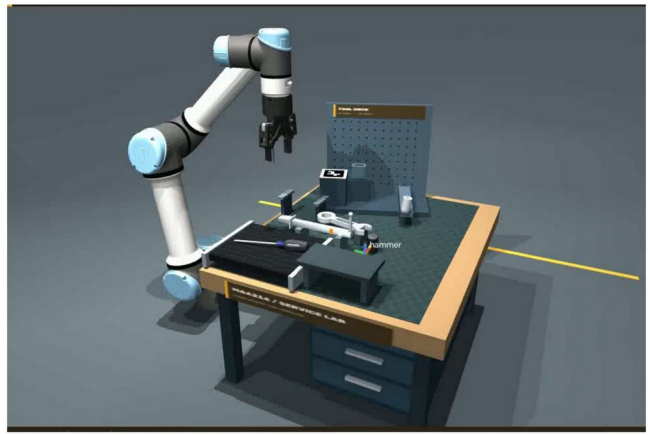
FINAL PROJECT 1

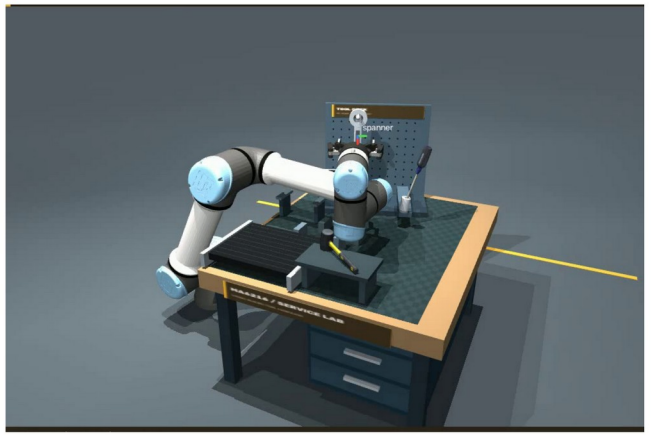
Calibrated 6D Vision for Sequential Tool Manipulation

a Initial workcell



t = 0.00 s | object pickup stations

b Released final state



t = 195.00 s | four acceptance checks passed

Independent calibration

6D pose-guided assembly

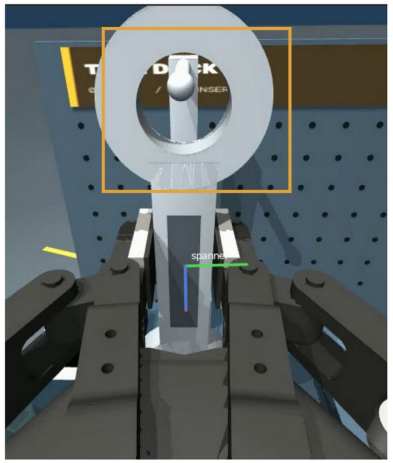
Controlled feedback comparison

A reproducible study of perception, physical manipulation and feedback.

UR5e + Robotiq 2F-85 | MuJoCo | Model-free reference reconstruction + FoundationPose

Research question and task constraints

a Precontact alignment



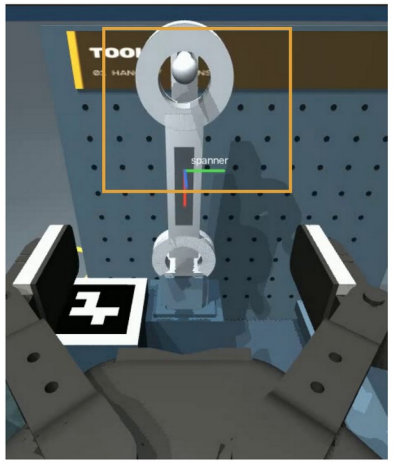
t = 171.00 s

b Hole-over-pin threading



t = 179.00 s

c Released hanging state



t = 195.00 s

Research question

Does execution-period 6D feedback improve complete-task reliability?

Geometric constraints

Tool-axis alignment

Movable fixture

Changed grasp transform

Local alignment and complete-sequence success are different endpoints.

System architecture and information boundaries

RGB-D
Estimated K

→

Localization
Foreground mask

→

6D pose
FoundationPose

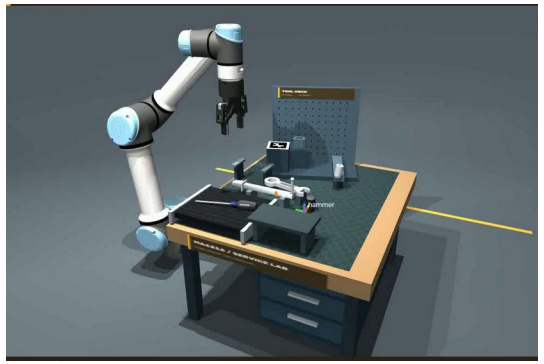
→

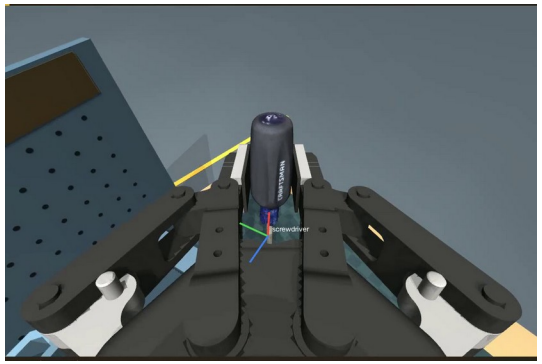
Robot target
Calibrated transforms

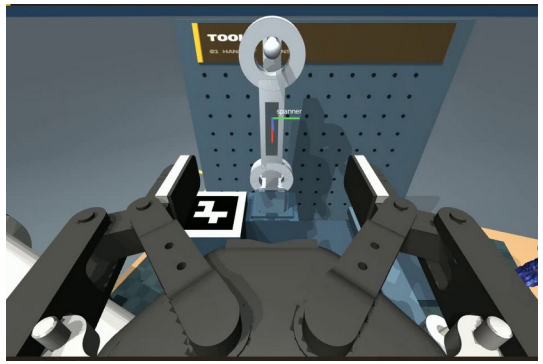
→

Execution
IK + joint control

Execution-period images → re-estimate pose → bounded target correction

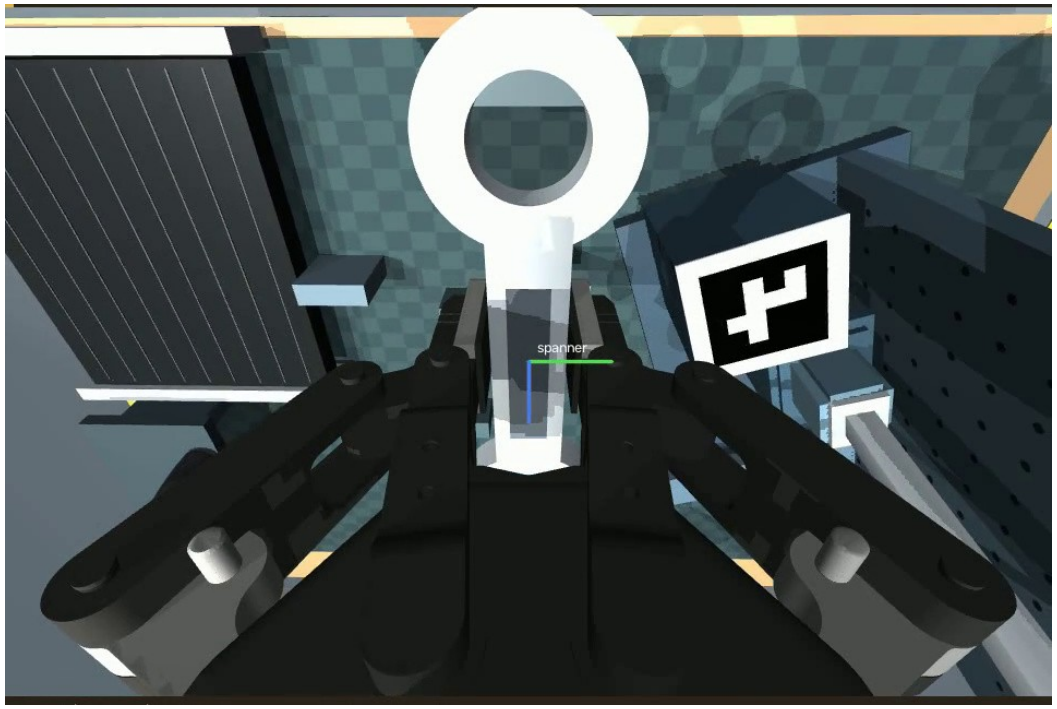






Simulation truth is used for scoring, not for commanded object poses.

Method selection for pose-guided manipulation



Selected: FoundationPose

Object-specific RGB-D reconstruction → camera-frame 6D pose

Reviewed alternatives

MegaPose: model-based 6D pose

GraspNet: grasp candidates, not assembly pose

“Model-free” describes the reference route; it does not mean reference-free.

Wen et al., CVPR 2024; Labbé et al., 2022; Fang et al., CVPR 2020; Todorov et al., IROS 2012

Image-based intrinsic and hand–eye calibration

a Checkerboard observations

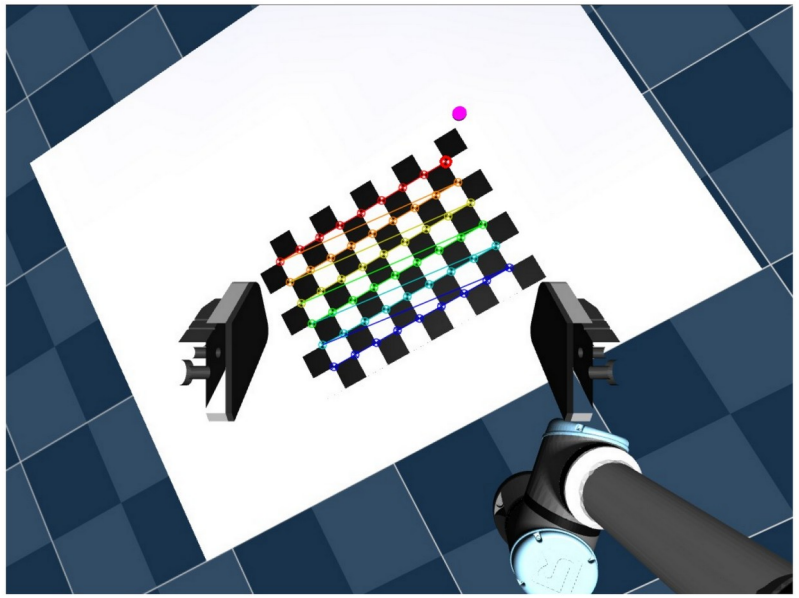
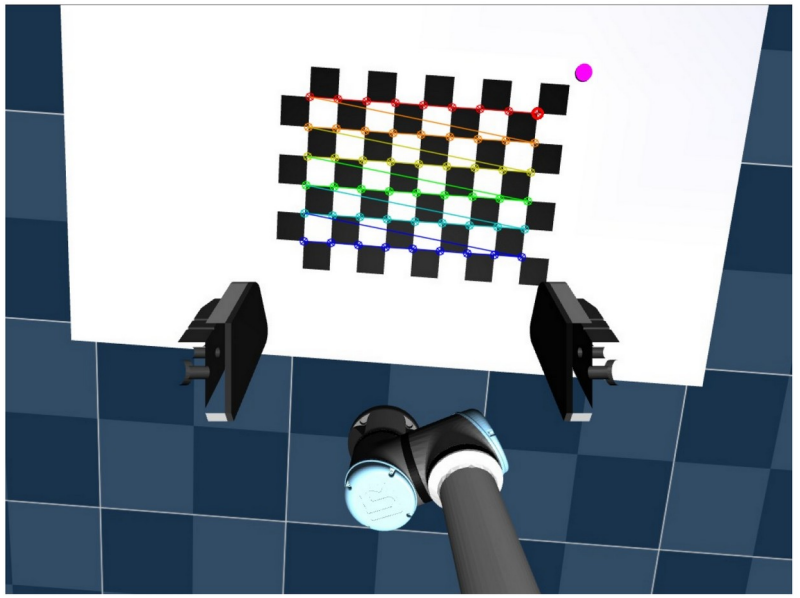


Image-based corner detection

b Held-out hand–eye view



PnP + robot FK; frozen validation chain

Intrinsic validation: 8 views. Hand–eye validation: 10 views. Ideal pinhole image formation.

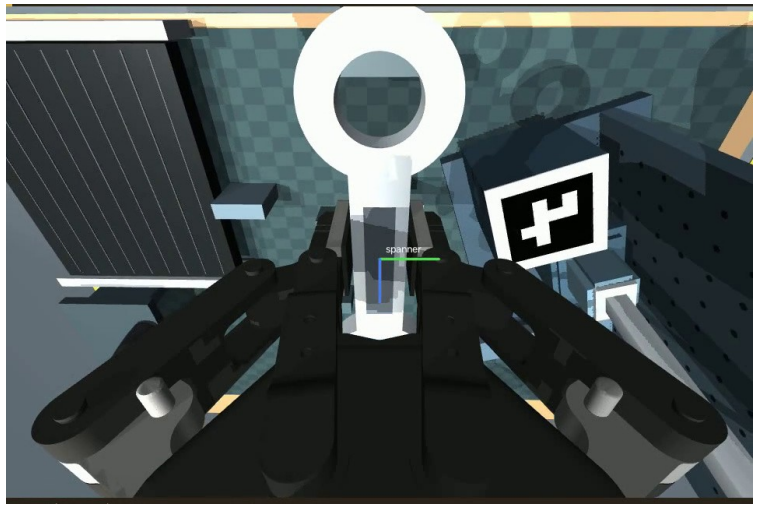
Wrist validation: 0.02177 px RMSE; hand–eye error 0.852 mm / 0.0227°.

Coordinate chain and the object-to-gripper transform

$$T_{BO} = T_{BG} \cdot T_{GC} \cdot T_{CO}$$

B: robot base G: gripper C: wrist camera O: object

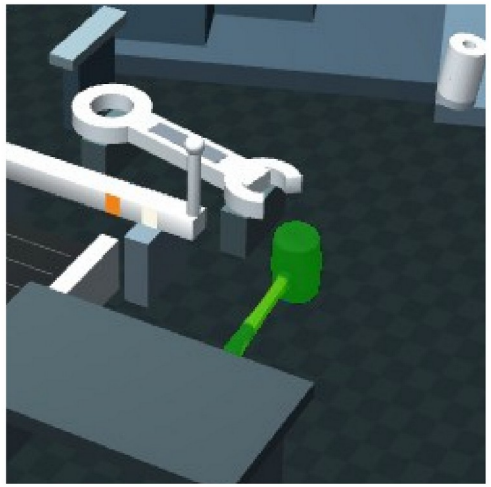




Regrasp requires an updated object-to-gripper transform.

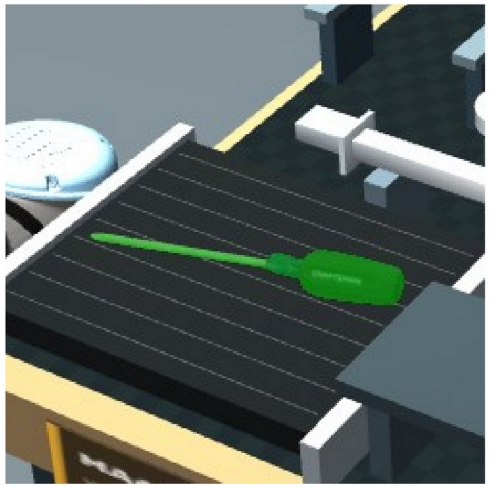
Localization outputs and the evaluation boundary

a Hammer



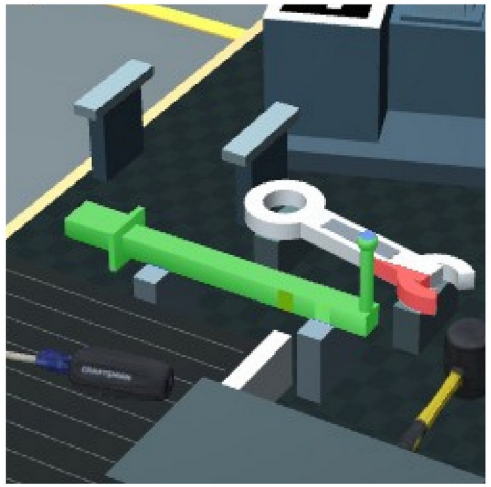
Nominal scene | seed 500

b Screwdriver



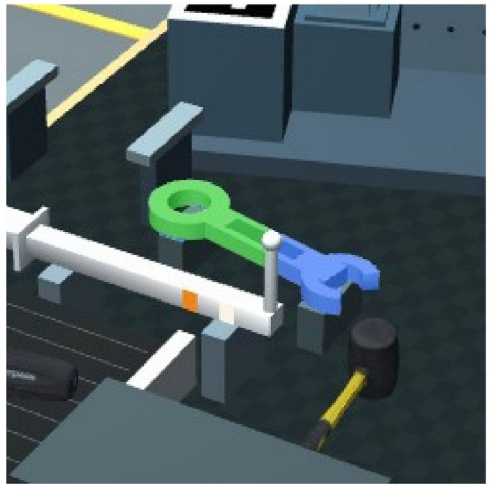
Nominal scene | seed 500

c Rack frame



Nominal scene | seed 500

d Spanner



Nominal scene | seed 500

Green: overlap; blue: missed target; red: extra foreground. 50 scenes / 200 queries.

Depth foreground masks are evaluated independently of manipulation success.

Discrete pose-based visual feedback before contact

$$e_p = p_{des} - p_{est} \qquad \Delta R = R_{des}\, R_{est}^{T}$$

Open loop

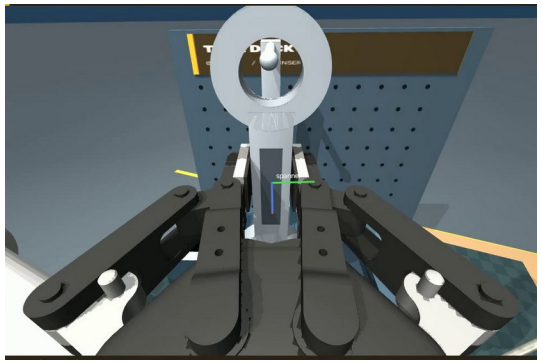
Initial images + feed prediction

Staged

Reobserve at task transitions

Iterative

Repeated bounded pose correction



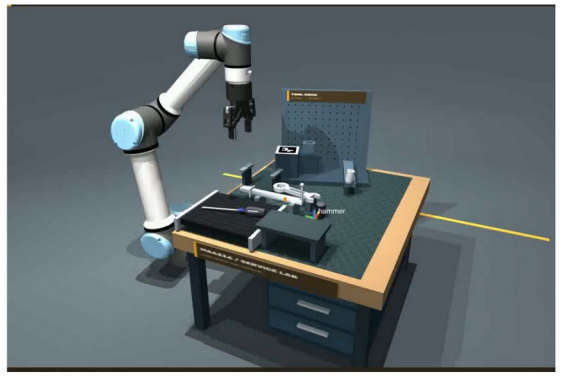
Per-step cap: 4 mm / 2°

Convergence: two observations within 1.5 mm / 1.5°

Visual residual ≠ true geometric error ≠ physical acceptance.

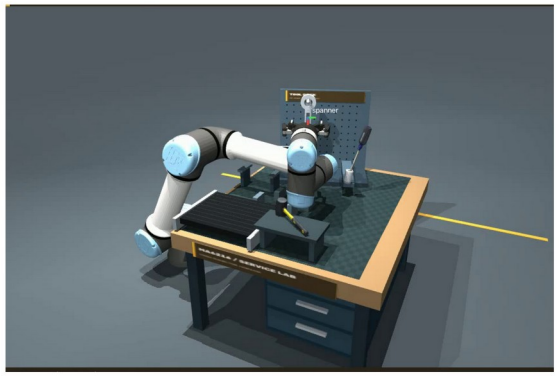
Paired experimental design and predefined endpoints

a Initial workcell



t = 0.00 s | object pickup stations

b Released final state



t = 195.00 s | four acceptance checks passed

3 × 2 × 8 = 48 attempts

Feedback mode × disturbance × seed

Same initial states within seed

Primary endpoint

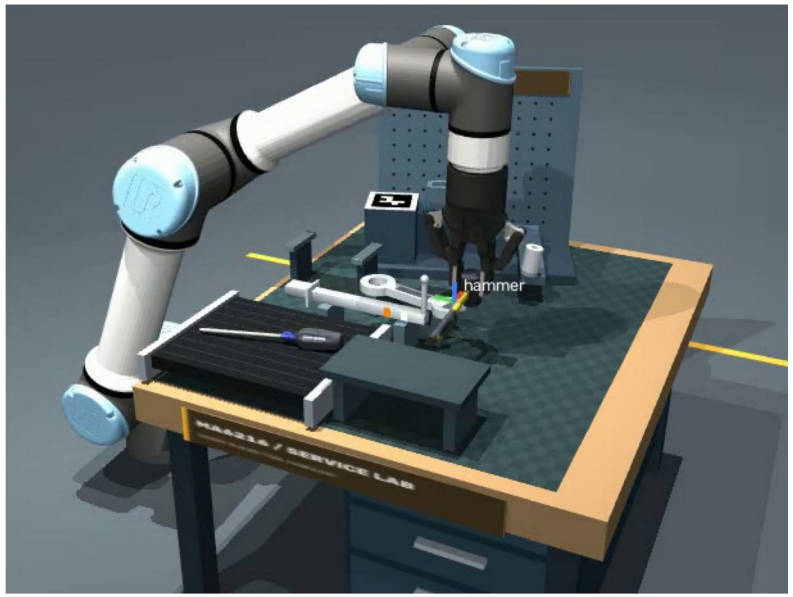
All four objects pass final geometry, support, release and stability checks.

Fixed: camera estimates, pose model, control thresholds, acceptance rules

All task failures are retained; reused seeds are explicitly disclosed.

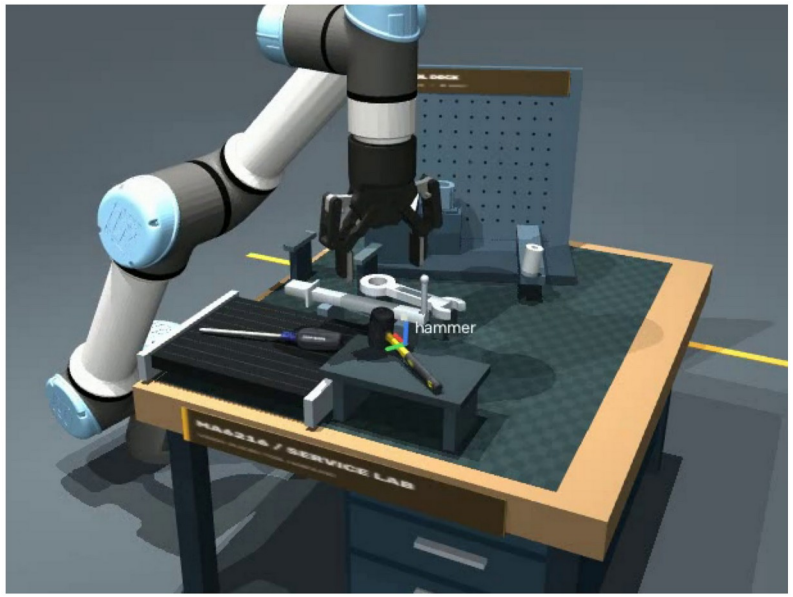
Physical tool return: grasp, transport and release

a Lift and transport



t = 13.60 s

b Released tray placement



t = 25.00 s

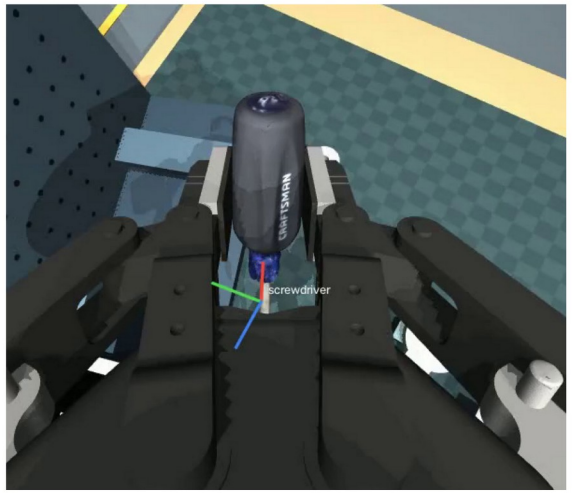
Spatially matched overview crops; no scene content or image contrast altered.

The acceptance criterion requires released support, not waypoint arrival.

Representative original run | iterative / no disturbance / seed 400 | playback time

Moving-target acquisition followed by tilted insertion

a Reorientation



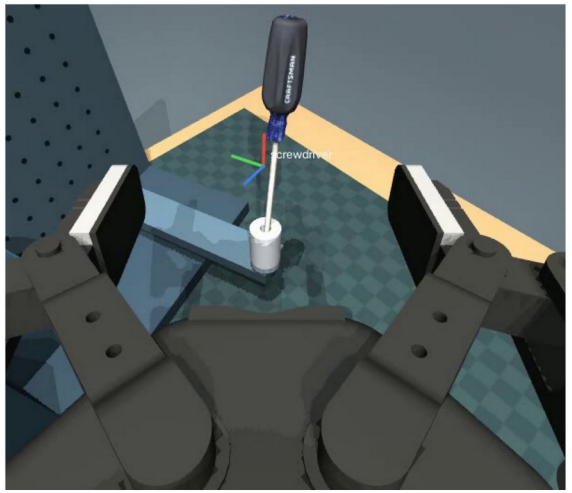
t = 50.00 s

b Guarded insertion



t = 60.00 s

c Released placement



t = 70.00 s

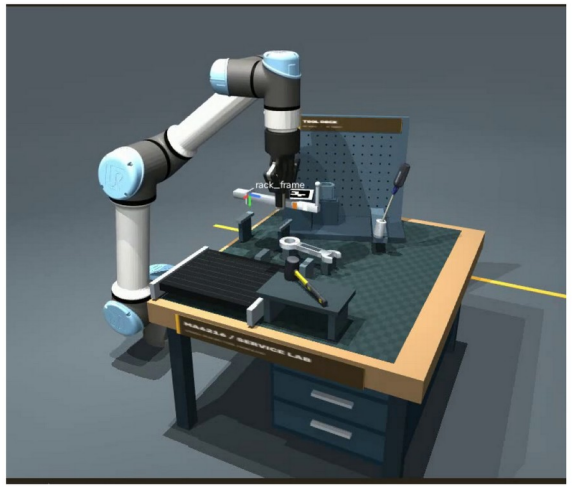
Wrist crops: reorientation → insertion → released placement. Playback times are not wall times.

The screwdriver shaft must align with the socket before guarded insertion.

Representative original run | iterative / no disturbance / seed 400 | playback time

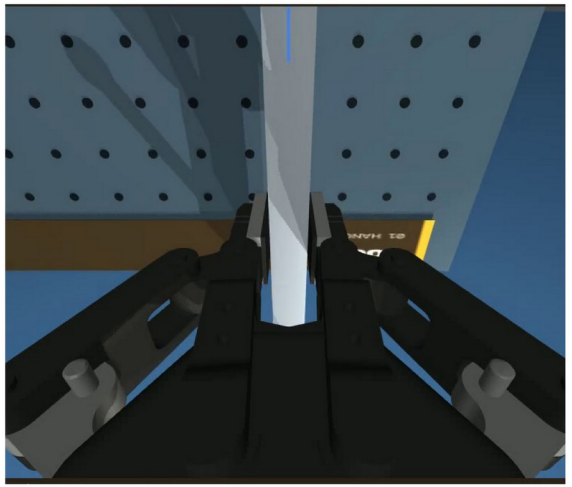
Rack installation creates a new downstream fixture

a Independent frame pickup



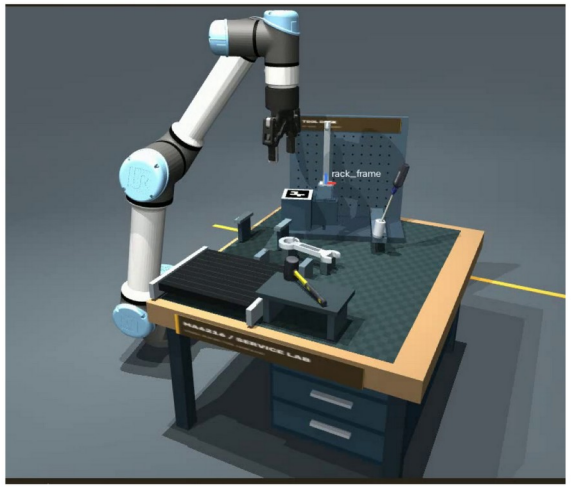
t = 91.00 s

b Frame–base seating



t = 112.00 s

c Released fixture



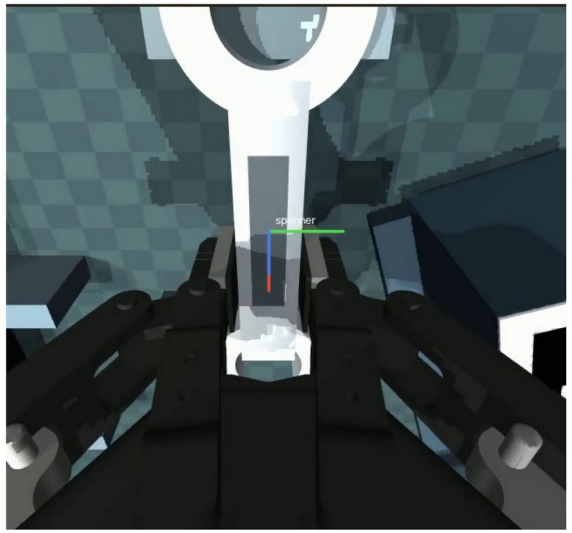
t = 134.40 s

The installed rack is released and relocalized before spanner hanging.

Representative original run | iterative / no disturbance / seed 400 | playback time

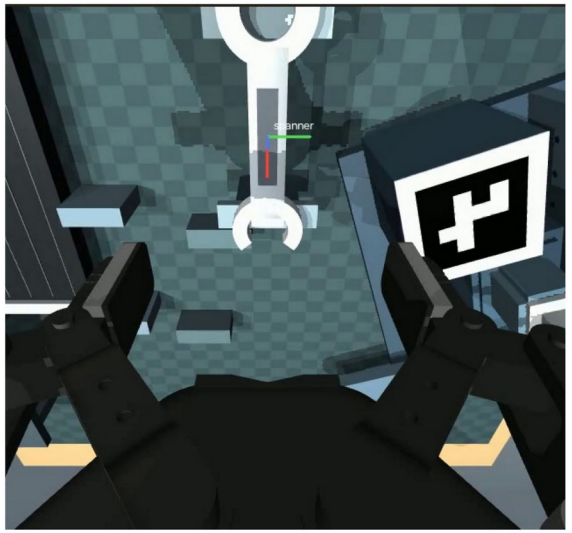
Physical regrasp: release, observe and reacquire

a Supported placement



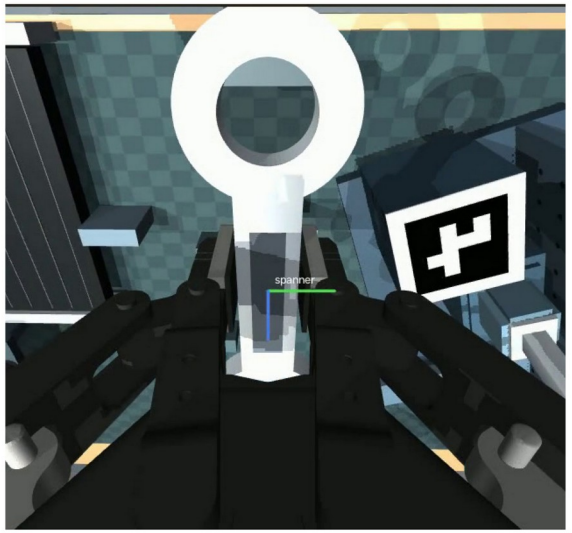
t = 147.00 s

b Complete finger clearance



t = 150.40 s

c New grasp after observation



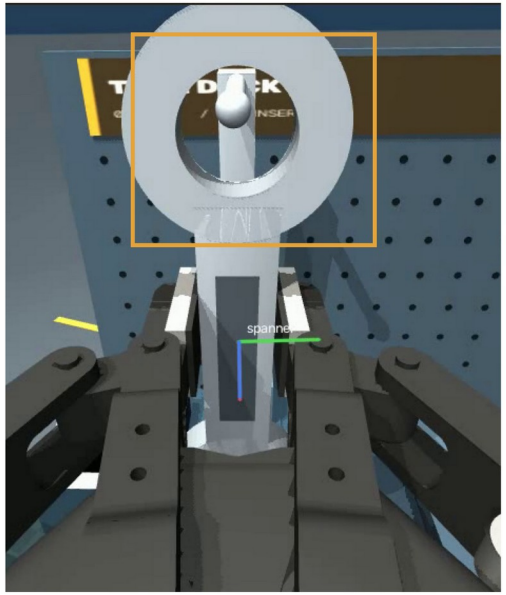
t = 158.20 s

Placement, finger clearance and reacquisition are all visible in the original run.

Representative original run | iterative / no disturbance / seed 400 | playback time

Hole-over-pin alignment and released support

a Precontact alignment



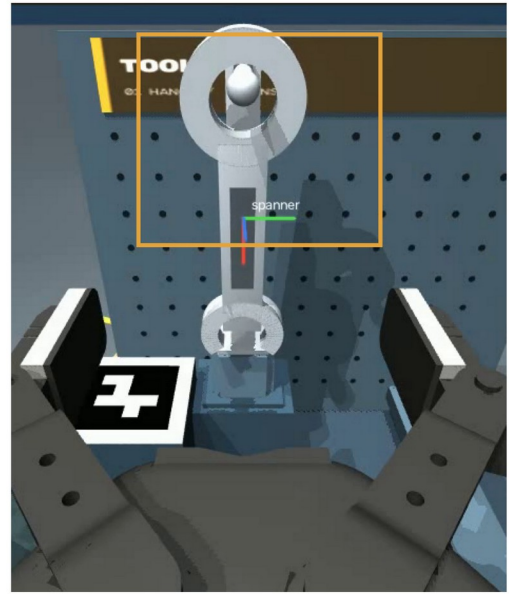
t = 171.00 s

b Hole-over-pin threading



t = 179.00 s

c Released hanging state



t = 195.00 s

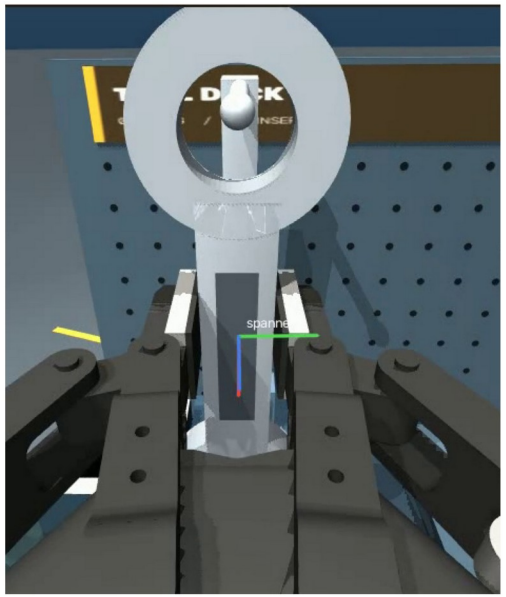
Amber rectangles identify the hole–pin region; annotations are explanatory, not measured axes.

Final success requires hanging support after gripper withdrawal.

Representative original run | iterative / no disturbance / seed 400 | playback time

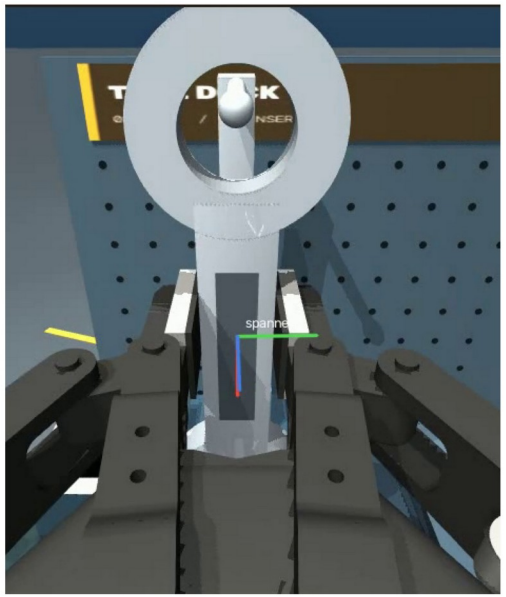
A logged correction in a perturbed integration run

a Before corrective action



Visual angular residual: 1.60°

b After correction and reobservation



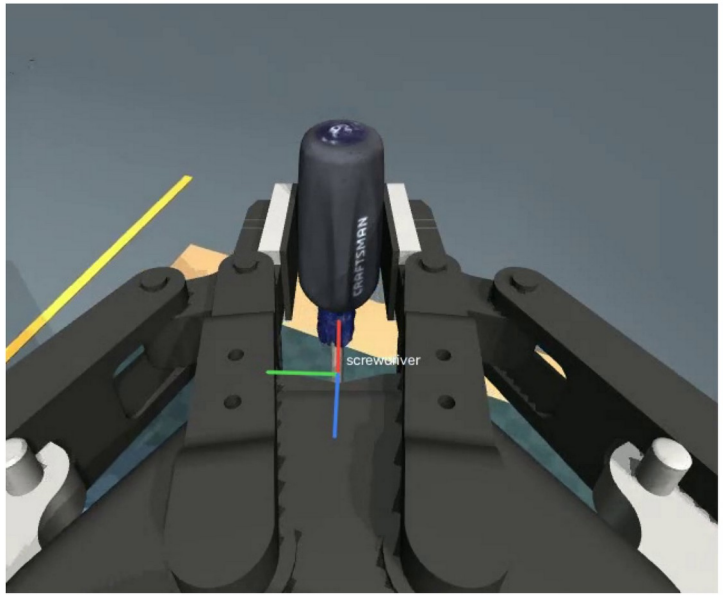
Visual angular residual: 0.74°

Separate integration run, seed 400; visual residuals only. Not added to the final 48 attempts.

One correction reduces the estimated angular residual in this example.

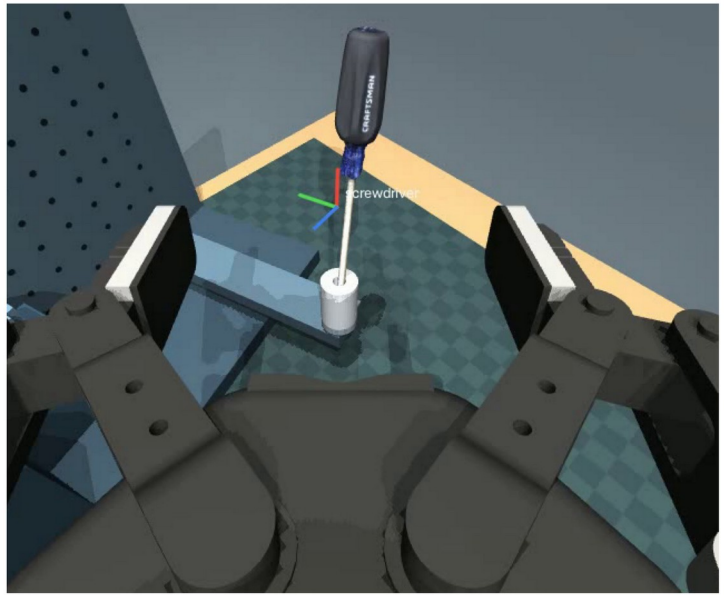
Paired insertion outcomes under the same initial state

a Open loop: guard stop



Seed 400 | t = 57.96 s

b Iterative: released driver



Seed 400 | t = 70.00 s

Same seed and initial state; different termination times. Failure reason taken from the original log.

This pair illustrates behavior; all attempted trials determine performance.

Matched failure: both methods stop before alignment

a Staged feedback



Seed 401 | contact guard at gripper closure

b Iterative feedback



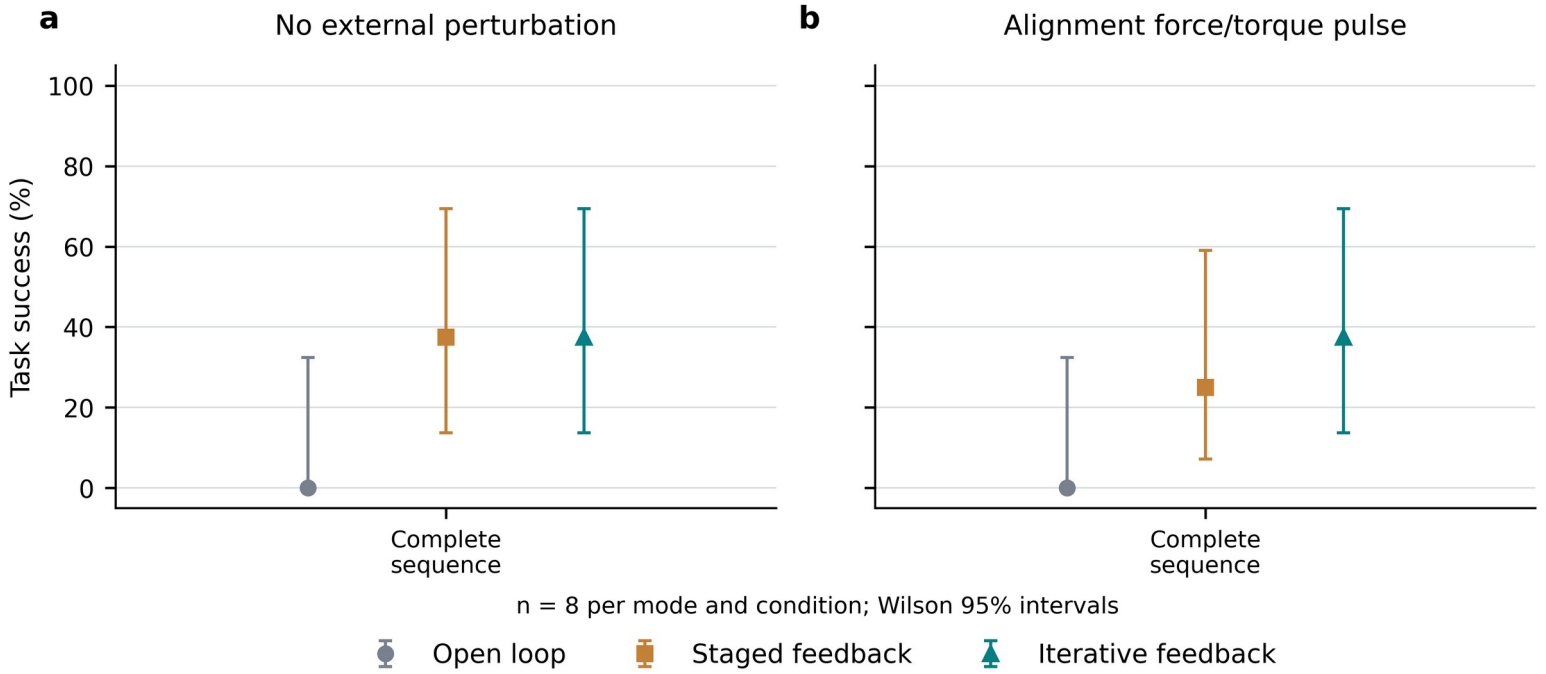
Seed 401 | same early failure stage

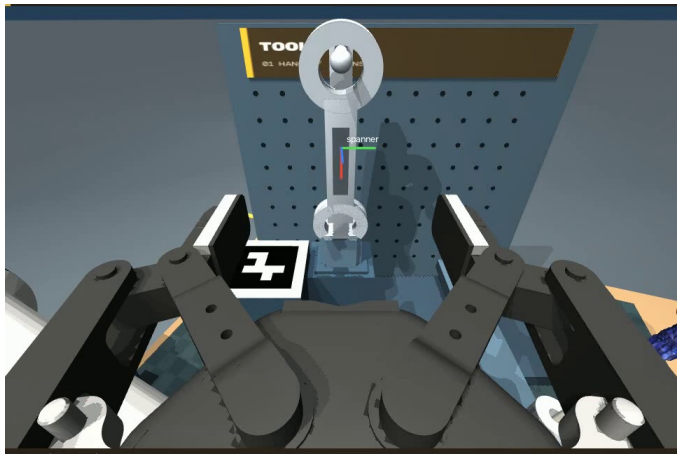
Both trials stop at CLOSE_GRIPPER. Similar appearance does not imply acceptable contact.

Early pickup reliability limits the benefit of downstream visual correction.

Final 48 attempts | staged vs iterative / no disturbance / seed 401

Complete-sequence success and uncertainty





11 / 48 complete successes

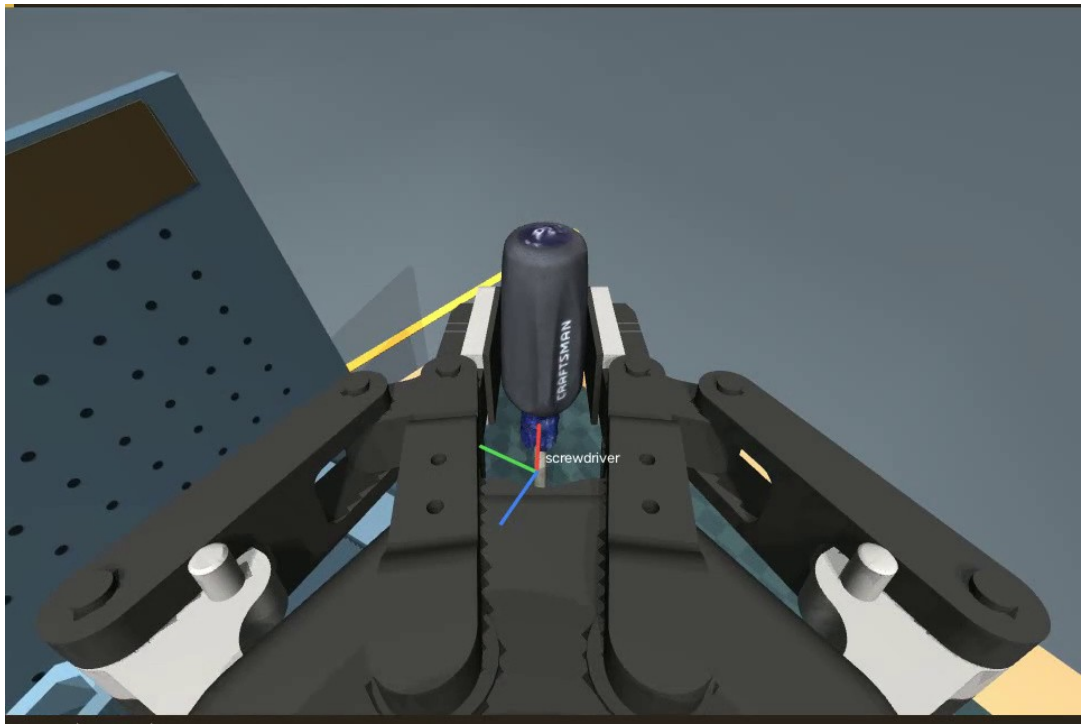
All failures retained

Adjusted p = 1.0

This sample does not establish a statistically significant feedback advantage.

n = 8 per mode × condition | Wilson 95% confidence intervals | exact paired tests with Holm correction

Pose accuracy and physical failure modes



Initial pose accuracy

Translation: 2.60 mm

Rotation: 1.44°

ADD / ADD-S: 3.17 / 2.29 mm

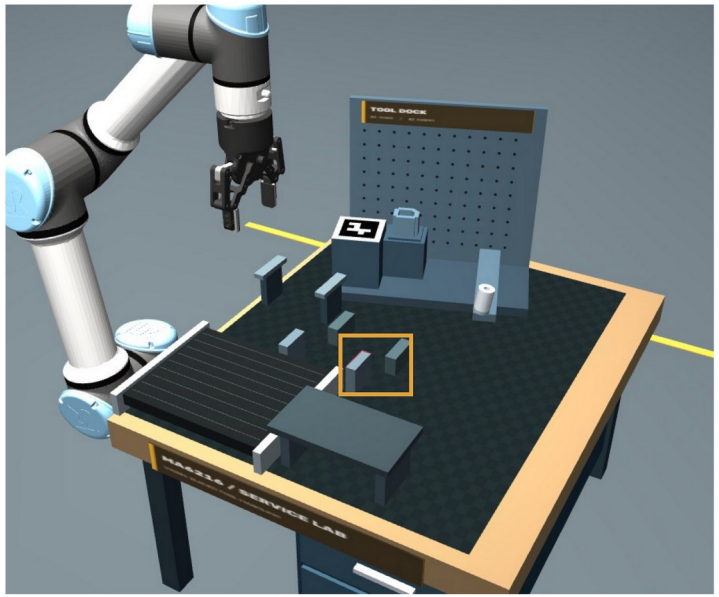
37 failures: 24 pickup · 6 hammer approach

6 open-loop insertion · 1 grasp loss

Accurate initial pose estimates do not guarantee reliable contact execution.

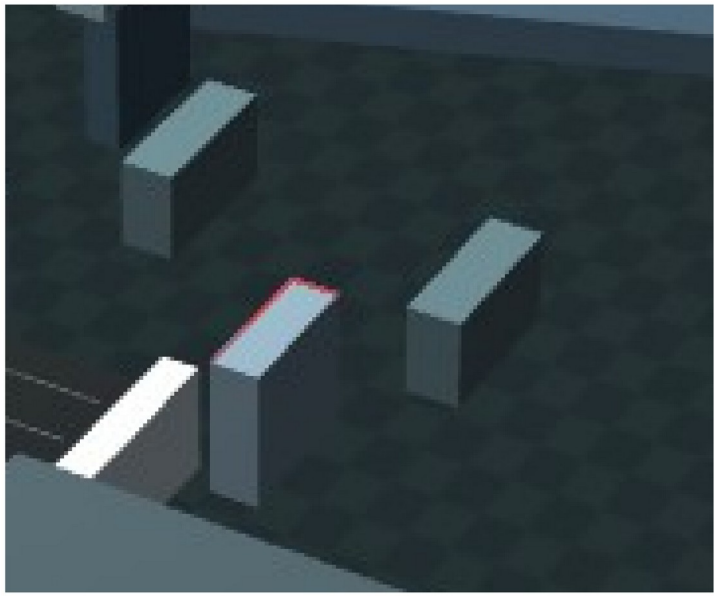
Background dependence bounds visual generalization

a Target-absent scene



Separate background-mismatch diagnosis

b Magnified false-positive region



Red pixels: predicted foreground without target

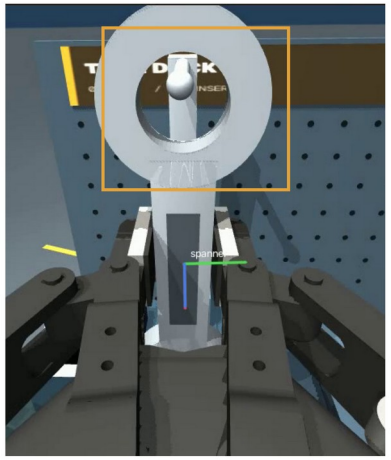
Context and magnified ROI from the same original image. Red: false-positive foreground.

A mismatched background creates a false-positive target region.

Separate localization diagnosis | target absent / rack frame / seed 500 | not a robot-trial outcome

Conclusions and next experiments

a Precontact alignment



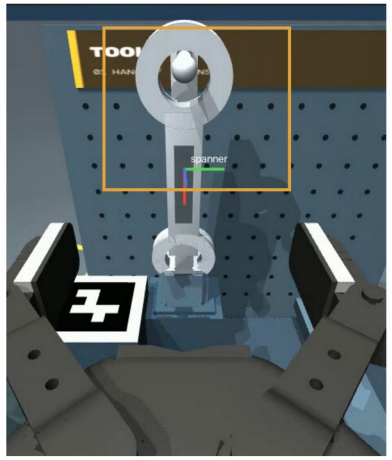
t = 171.00 s

b Hole-over-pin threading



t = 179.00 s

c Released hanging state



t = 195.00 s

Supported

Calibrated perception

Physical task integration

Not established

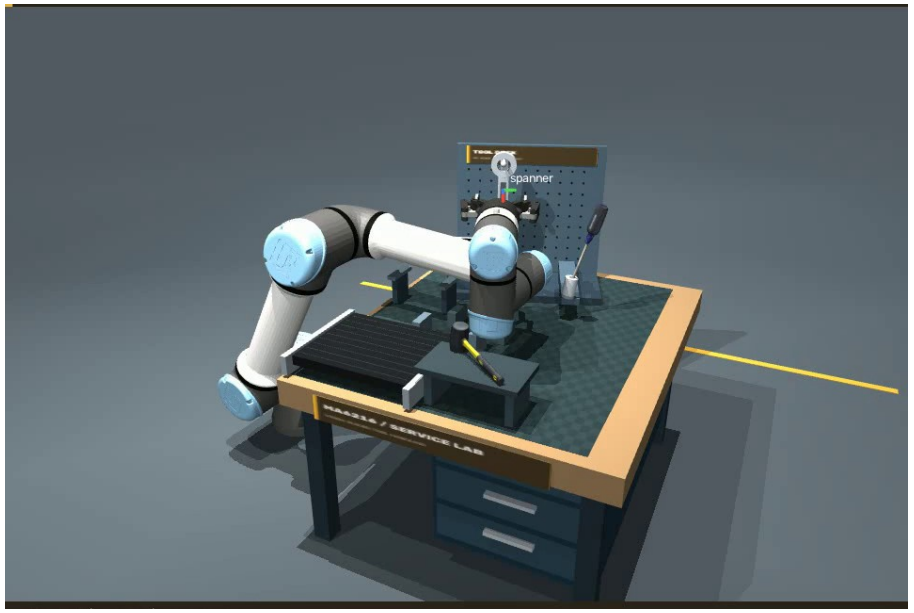
Architectural superiority

Unseen-scene generalization

Real-robot reliability

Priority: pickup reliability, then broader controlled evaluation.

Primary references and reproducible evidence



Wen et al. FoundationPose. CVPR, 2024.

Zhang. A flexible new technique for camera calibration. TPAMI, 2000.

OpenCV. Camera and hand–eye calibration documentation.

Chaumette & Hutchinson. Visual servo control, Part I. RAM, 2006.

Todorov et al. MuJoCo. IROS, 2012.

Labbé et al. MegaPose. arXiv:2212.06870, 2022.

Fang et al. GraspNet-1Billion. CVPR, 2020.

Code + data + original images + source hashes support independent review.

Full citations: docs/LITERATURE_AND_SELECTION.md and docs/references.bib | source manifests included